Indian Institute of Technology (BHU) Varanasi
End-Sem Examination (Odd Sem 2025-26)
MO 201 Materials Science

Time: 3 hours **Full marks: 50**

There are 12 questions in this question paper. Attempt any 10 questions. All questions carry equal marks. Do not attempt extra questions. Only your first 10 answers will be considered, not the best 10. Non-programmable scientific calculator is allowed.

Q1. Explain (a) Shockley and (b) Frank partial dislocations in FCC. (c) Write the dislocation reaction for Shockley partial dislocations in HCP. [2 + 2 + 1 = 5]

Q2. (a) Explain ductile to brittle transition in iron and Yoffee diagram. (b) Explain with diagram the formation of a sessile dislocation in an FCC crystal by a dislocation reaction. [3 + 2 = 5]

Q3. Explain (a) Brinell, Rockwell, Vickers hardness tests (b) 3-point bending test. [3 + 2 = 5]

Q4. What are (a) cermets and (b) hybrid composites? (c) What is the importance of critical fiber length? [2+2 +1=5]

Q5. (a) Why is carbon black used and not silica as particle reinforcement in vulcanized rubber? (b) Explain the types of structural composites. [1 + 4 = 5]

Q6. Draw the iron-carbon phase diagram showing all the phases and invariant reactions. [5]

Q7. Draw the Ag-Pt and Pb-Sn phase diagrams and explain the invariant reactions in each. What is Lever rule? [2 + 2 + 1 = 5]

Q8. Explain the free electron theory and on the basis of the same, explain why solids conduct electricity. [5]

Q9. Explain (a) intrinsic semiconductor (b) p-type extrinsic semiconductor (c) n-type extrinsic semiconductor. [1+2+2 =5]

Q10. Explain (a) variation of conductivity with temperature for an extrinsic semiconductor (b) forward and reverse bias in junction diode. [3+2 =5]

Q11. (a) Explain ferromagnetism in Heusler alloys although Mn is not ferromagnetic in elemental form. (b) Explain what happens to the magnetization of $NiFe_2O_4$ upon addition of Zn. (c) What is ferrimagnetism? [2+2+1=5]

Q12. (a) Define the various polarization mechanisms. (b) Explain why $BaTiO_3$ shows ferroelectricity. [2+3=5]

****